

Course: B. Tech.

Branch : Civil

Semester : VII

Subject Code & Name:BTCVC703&Water Resources Engineering

Max Marks: 60

Date:01/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

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|---|------|---|
| A) Write the Necessity and ill effects of Irrigation | CO 1 | 6 |
| B) Explain with neat sketch Hydrological Cycle | CO 1 | 6 |
| C) Compute the depth and frequency of irrigation required for a certain crop with the data given below. | CO 2 | 6 |

Root zone depth = 100cm

Field capacity = 22%

Wilting point = 12%

Dry unit weight of soil = 14.78 kN/m³

Consumptive use = 25mm/day

Efficiency of irrigation = 50%

Assume 50% depletion on available moisture before application of irrigation water at field capacity.

Q.2 Solve Any Two of the following. 12

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|---|------|---|
| A) Write the design criteria for Gravity Dam | CO 3 | 6 |
| B) Write the different modes of failure of Earthen Dam | CO 2 | 6 |
| C) Define duty and delta and Relation between Duty and Delta. | CO 1 | 6 |

Q. 3 Solve Any Two of the following. 12

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|--|------|---|
| A) Write the Necessity and Different Types spillway in details | CO 2 | 6 |
| B) The isohyetal map for 24 hours storm gave area enclosed between different isohyets as follows | CO 1 | 6 |

Isohyets in mm	21	20	19	18	17	16	15	14	13	12
Enclosed Area sq km	543	1345	2030	2545	2955	3280	3535	3710	3880	3915

Determine the average depth of rainfall over the basin

C) The following are rates of rainfall for successive 20 minutes period of 140 minutes storm 2.5, 2.5, 10.0, 7.5, 1.25, 1.25, 5.0 cm/hr taking the value of Phi Index as 3.2 cm / hr find out net runoff in cm the total rainfall and value of W index	CO 1	6
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Q.4 Solve Any Two of the following.		12
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|---|------|---|
| A) Write a note on 1) Unit Hydrograph 2) S hydrograph | CO 2 | 6 |
| B) Explain with neat sketch different types of Tube well | CO 3 | 6 |
| C) Explain Bligh's Creep Theory and Limitations of Bligh's Creep Theory | CO 3 | 6 |

Q. 5 Solve Any Two of the following.		12
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|---|------|---|
| A) What are the different causes of Water Logging also suggest their Preventive measures. | CO 1 | 6 |
| B) Explain with neat sketch Various components and their design principles of Lift irrigation schemes | CO 2 | 6 |
| C) Write the different techniques for Rain water Harvesting | CO 3 | 6 |

*** End ***